Rice Leaf Disease Detection in the Philippines using EfficientNet-based Convolutional Neural Networks with Multi-Metric Performance Evaluation

Francine Angela G. Alejandro
School of Computing,
Holy Angel University
Angeles City, Pampanga, Philippines
+639287207896
cinella.alejandro07@gmail.com

Yohanna A. Decilio
School of Computing,
Holy Angel University
Angeles City, Pampanga, Philippines
+639629393821
yohannadecilio@gmail.com

Shane S. Mendoza
School of Computing
Holy Angel University
Angeles City, Pampanga, Philippines
+639310482585
mendozashane0820@gmail.com

Maria Niña Grace L. Pascua
School of Computing
Holy Angel University
Angeles City, Pampanga, Philippines
+639150547753
ninapascua88@gmail.com

ABSTRACT

Rice is a staple crop that plays a crucial role in food security and agricultural livelihoods, particularly in the Philippines. However, rice production is often affected by leaf diseases such as Rice Blast, Bacterial Leaf Blight, Brown Spot, Sheath Blight, and Tungro, which can significantly reduce crop yields by 20%–70% during severe outbreaks. Identifying these diseases typically relies on the experience and observation of farmers and agricultural specialists. While this expertise is valuable, timely large-scale screening can still be difficult, especially when symptoms appear similar across diseases, early signs are subtle, or expert support is not immediately available. This project proposes a rice leaf disease detection system using an EfficientNet-based Convolutional Neural Network (CNN) for image classification. The system allows users to upload rice leaf images and obtain the predicted disease, its confidence score, and a short explanation of the result. Model performance will be evaluated using accuracy, macro-F1 score, and a per-class confusion matrix will be used for class-level error analysis, while Grad-CAM visualization will be used to highlight the image regions influencing model classification and improve interpretability. To improve interpretability, an auxiliary Natural Language Processing (NLP) component retrieves short explanations of the detected disease using TF-IDF similarity matching and displays them alongside the predicted disease and confidence score. For development purposes, model evaluation metrics will also be shown. In addition, a reinforcement learning (RL) component uses a simple Q-learning agent to tune the classification confidence threshold based on improvements in the validation macro-F1 score. The system is designed to assist farmers and agricultural specialists in crop monitoring.

CCS Concepts

- **Computing Methodologies** → Machine Learning;
- **Applied Computing** → Agriculture;
- **Computing Methodologies** → Neural Networks

Keywords

Rice leaf disease detection; EfficientNetB0; Grad-CAM; NLP advisory system; reinforcement learning; plant disease classification; Pampanga; food security

1. INTRODUCTION

Rice is the primary staple crop of the Philippines and accounts for around 20% of the gross value added of Philippine agriculture [1], while the rice sector supports about 2.5 million households [2]. This significance is mirrored in Pampanga, a major rice-producing province in Central Luzon, where the crop holds deep economic and cultural significance. However, despite its importance, rice leaf diseases remain a persistent threat to yield stability and farmer income.

Experienced Filipino rice farmers possess deep, field-based knowledge of their land and crops—many can identify disease signs from subtle changes in leaf color, texture, and growth pattern long before visible lesions fully develop. Even so, accurate identification can remain difficult when multiple diseases appear simultaneously or when symptoms present atypically under varying field conditions. Five diseases are particularly prevalent in Philippine paddy fields: Rice Blast (*Magnaporthe oryzae*), Bacterial Leaf Blight (*Xanthomonas oryzae*), Brown Spot (*Bipolaris oryzae*), Sheath Blight (*Rhizoctonia solani*), and Tungro. Collectively, these diseases can substantially reduce rice yield under severe outbreak conditions [3].

The five diseases exhibit strong inter-class visual similarity and wide intra-class variability driven by cultivar, growth stage, and field environment. Images collected in real paddy fields introduce additional variation from changing light conditions, background clutter, moisture, and co-occurring insect damage [3]. These factors make even well-trained classifiers prone to error in boundary cases.

This project proposes a rice leaf disease detection system: a fine-tuned EfficientNetB0 convolutional neural network that classifies six classes from field photographs and generates Grad-CAM saliency maps, enabling inspection and verification of

each classification. The system also includes an auxiliary Natural Language Processing (NLP) component that retrieves short explanations describing the predicted rice disease using TF-IDF similarity matching from a small disease knowledge base. In addition, a reinforcement learning component is implemented to tune the classification confidence threshold using a simple Q-learning agent that learns a threshold policy based on improvements in validation macro-F1 score. The rice leaf disease detection is designed as a reference tool that places additional evidence in the farmer's hands, supporting—not replacing—the judgment they have built through years of working the land.

2. PROBLEM STATEMENT AND SCOPE

The study addresses the challenge of accurately and consistently identifying rice leaf diseases under Philippine field conditions, where visually similar symptoms, environmental variation, and limited immediate technical support can make diagnosis difficult. The project aims to develop an EfficientNet-based rice leaf disease detection system that is accurate, interpretable, and usable as a decision support tool for farmers and agricultural specialists.

2.1 Scope

This project focuses on the development of a rice leaf disease detection system that classifies images of rice leaves into six classes: the five common diseases affecting Philippine rice fields (Rice Blast, Bacterial Leaf Blight, Brown Spot, Sheath Blight, and Tungro) and a Healthy class. The system uses an EfficientNet-based Convolutional Neural Network (CNN) to analyze uploaded leaf images and classify the corresponding disease category with a confidence score.

The study includes dataset preparation, image preprocessing, model training, and evaluation using performance metrics such as accuracy, macro-F1 score, and per-class confusion matrix. Grad-CAM visualization will be implemented to highlight the regions of the leaf image that influenced the model's classification, improving interpretability for users.

In addition to the CNN-based classification component, an auxiliary NLP module retrieves short explanations of the detected disease from a small knowledge base using TF-IDF similarity matching and displays them with the predicted disease and confidence score. For development purposes, model evaluation metrics such as accuracy, macro-F1 score, and the per-class confusion matrix will also be shown. A reinforcement learning component using a simple Q-learning agent will also be implemented to tune the classification confidence threshold based on the validation macro-F1 score.

The system is designed as a decision support tool to assist farmers and agricultural specialists in crop monitoring and disease detection. However, the project is limited to image-based classification and does not include real-time field deployment, treatment recommendation systems, or integration with mobile or agricultural management platforms. Model performance may also be influenced by variations in lighting conditions, leaf appearance, and dataset quality.

2.2 Success Metrics and Constraints

2.2.1 Success Metrics

The performance of the CNN model will be evaluated using key classification metrics to assess classification reliability and class-level performance. The primary evaluation metrics include accuracy, macro-F1 score, and per-class confusion matrix. Accuracy measures the overall proportion of correctly classified rice leaf images. The macro-F1 score evaluates model performance across all disease classes by equally considering each class in the dataset, which is important for multiclass disease classification. The per-class confusion matrix provides a detailed breakdown of classification results, allowing analysis of how frequently specific diseases are correctly classified or confused with other classes.

2.2.2 Constraints

Several constraints may influence the implementation of the project. First, publicly available rice leaf image datasets may contain class imbalance, limited environmental variation, or inconsistent image quality. Second, agricultural images often contain background noise such as soil particles, water droplets, insect damage, or variable lighting conditions that can affect model training. Finally, computational limitations and project time constraints may limit the number of model architectures or hyperparameter configurations that can be explored.

3. RELATED WORK

CNNs have established a strong record in plant disease recognition. Mohanty et al. [4] showed that deep networks trained on the PlantVillage dataset could identify 26 diseases across 14 crops with over 99% laboratory accuracy, though performance fell significantly on field images—highlighting the domain gap that motivates the augmentation strategies used in this work. This finding is important because it shows that strong results in controlled datasets do not always guarantee the same level of performance in real agricultural environments. In practical field conditions, leaf images may contain uneven lighting, blurred details, overlapping leaves, and noisy backgrounds, all of which can reduce classification accuracy. As a result, plant disease detection systems must be designed not only for high laboratory performance but also for strong adaptability to real-world conditions. Chin and Suhaili [5] highlight inter-class similarity as a challenge in fine-grained paddy disease classification. This challenge becomes even more significant in rice disease detection because several diseases share similar discoloration patterns, lesion shapes, or early visual symptoms. Such visual overlap makes accurate classification difficult and increases the need for a model that can extract more discriminative and fine-grained image features.

Tan and Le [6] introduced EfficientNet, which applies compound scaling to network depth, width, and input resolution simultaneously. EfficientNetB0 matches the accuracy of much larger models at a fraction of the parameter count, making it well-suited to web or edge deployment scenarios where inference speed and model size matter. This efficiency motivates its selection as the backbone for rice leaf disease detection.

Mumuni and Mumuni [7] surveyed data augmentation strategies, demonstrating that geometric and photometric transforms—rotation, brightness jitter, contrast adjustment, and horizontal flipping—substantially improve generalization when training images are environmentally homogenous. Buda et al. [8] showed that class imbalance suppresses minority-class recall in CNNs and that macro-F1 is a more honest performance indicator for rice leaf disease detection's training and evaluation design.

Sarkar [9] showed that Grad-CAM can improve interpretability in rice disease classification by highlighting lesion regions relevant to the model's prediction. This finding supports Grad-CAM as an interpretability feature in rice leaf disease detection.

4. DATASET PLAN

4.1 Sources and Licensing

The dataset comprises publicly available rice leaf image datasets from Kaggle, including the Rice Disease Dataset [10], the Rice Leafs Dataset [11], and the Philippines Rice Diseases [12]. Images will be filtered and labeled to match the six classes used in the study. These datasets are used in accordance with their respective licenses provided on the Kaggle Platform. Healthy leaf samples are included for comparison. No identifiable human subjects or private land data are collected or stored.

4.2 Labels and Representativeness

Six (6) classes are defined: Rice Blast, Bacterial Leaf Blight, Brown Spot, Sheath Blight, Tungro, and Healthy. The dataset is intended to include variation in lighting conditions, leaf orientations, and field conditions to better reflect real-world agriculture environments, consistent with challenges identified in rice leaf disease literature [3].

5. METHODOLOGY

5.1 Convolutional Neural Network (CNN) Component

The primary classification component uses EfficientNetB0 [6] with ImageNet pre-trained weights via transfer learning. Data augmentation techniques, including rotation, brightness and contrast adjustments, and flipping, are applied [7]. Grad-CAM visual explanations [13] offer interpretability by highlighting the image regions most influential to each prediction. Evaluation metrics include accuracy, macro-F1 score, and per-class confusion matrix.

5.2 Natural Language Processing Component

An auxiliary Natural Language Processing (NLP) component will be implemented to retrieve and generate short explanations describing the detected rice disease. The system uses a small disease knowledge base containing descriptions of symptoms and characteristics for each disease class. After classification, the NLP module retrieves the relevant description using TF-IDF similarity matching and presents it as a concise explanation accompanying the predicted disease and confidence score. For development and evaluation purposes, additional model performance metrics such as accuracy, macro-F1 score, and the per-class confusion matrix will also be displayed to provide insight into the system's performance. This allows users to better understand the model's prediction through natural language descriptions.

5.3 Reinforcement Learning Component

A reinforcement learning component will be implemented to tune the classification confidence threshold using a simple Q-learning agent [14]. The agent interacts with a validation environment where actions adjust the prediction threshold applied to model outputs. Rewards are computed based on improvements in the macro-F1 score on the validation dataset. Through this interaction, the agent learns a threshold policy that maximizes classification performance.

5.4 Minimum Viable Product (MVP)

The minimum viable product implements the core functionality of the rice leaf disease detection system using the EfficientNetB0-based CNN model for image classification. Users will be able to upload rice leaf images through a web interface and receive the predicted disease category along with a confidence score. Grad-CAM visualization will highlight the regions of the leaf image that most influenced the model's prediction, while a short explanation retrieved by the NLP component from a small disease knowledge base will describe the disease alongside the predicted class and confidence score.

6. ETHICS RISK REGISTER

6.1 Misdiagnosis Leading to Crop Loss

CNN classification errors may lead to misdiagnoses, which could contribute to inappropriate disease management decisions and potentially affect crop yield [3]. To mitigate this, the system displays confidence scores and Grad-CAM heatmaps for every prediction. All system outputs carry a clear disclaimer encouraging users to confirm any treatment decision with a licensed agricultural technician or extension officer.

6.2 Dataset Bias from Class Imbalance

Some disease classes may be underrepresented in publicly available datasets, which can reduce performance on minority classes [8]. To mitigate this risk, balanced data augmentation is applied to underrepresented classes, while macro-F1 score and per-class confusion matrices are used to evaluate performance and ensure fairness across all classes.

6.3 Overreliance on Automated Recommendations

Users may depend too heavily on the system's output without considering specific field conditions or unique farm situations. The system is meant to support decision-making, not replace expert judgment.

7. TEAM ROLES AND MILESTONES

7.1 Team Roles

- Francine A. G. Alejandro - Project & Integration Lead: Defines scope, manages timeline, oversees component integration, handles release tagging, coordinates defense.
- Yohanna A. Decilio - Data & Ethics Lead: Sources datasets, manages licensing and consent, conducts bias checks, writes ethics statements.
- Shane S. Mendoza - Modeling Lead (CNN/NLP): Designs models, develops training pipelines, conducts ablation studies, performs error analysis, prepares evaluation metrics, and reports.
- Maria N. G. L. Pascua - Documentation & MLOps Lead: Sets up repository, tracks issues, maintains project board, ensures reproducibility, prepares environment files, and automates scripts/CI.

7.2 Milestones

Week 1

- Project kickoff; team roles assigned
- Dataset exploration; initial data acquisition
- Repository setup; GitHub board created
- Initial CNN and NLP research; literature review

Week 2

- Data acquired and cleaned; train/validation/test splits finalized
- Baseline models trained (simple ML and DL); initial metrics logged
- CNN experiment running with first results; NLP component scaffolded/prototyped

- RL agent stubbed with reward design; early learning curves prepared
- Draft model card sections started; Ethics statement updated
- Deliverable: 1-2 page Checkpoint Report and updated repository

Week 3

- Code frozen and reproducible
- Results tables/plots generated; ablations (≥2); error/slice analysis completed
- Final Report (4-6 pages) in conference-style PDF format
- Slide Deck (8-10 slides) and Live Demo
- Defense: 8-10 minutes presentation and Q&A

8. REFERENCES

[1] Philippine Statistics Authority. 2025. Agriculture and Fisheries Indicators System: Tracking Economic Growth, 2020–2024. Agriculture and Fisheries Indicators System. *Psa.gov.ph*. Retrieved from https://psa.gov.ph/system/files/main-publication/AFIS%20Economic%20Growth%20Agriculture%202020-2024_signed.pdf

[2] Philippine Statistics Authority. 2023. Crops Production Survey 2020. *psada.psa.gov.ph*. Retrieved from https://psada.psa.gov.ph/catalog/234

[3] Chinna Gopi Simhadri, Hari Kishan Kondaveeti, Valli Kumari Vatsavayi, Alakananda Mitra, and Preethi Ananthachari. 2024. Deep learning for rice leaf disease detection: A systematic literature review on emerging trends, methodologies, and techniques. Information processing in agriculture (May 2024). DOI:https://doi.org/10.1016/j.inpa.2024.04.006

[4] Sharada P. Mohanty, David P. Hughes, and Marcel Salathé. 2016. Using Deep Learning for Image-Based Plant Disease Detection. *Frontiers in Plant Science* 7, (September 2016). DOI:https://doi.org/10.3389/fpls.2016.01419

[5] Wallace F. L. Chin and Wida S Suhaili. 2025. Class-Specific Cross Attention for Fine-Grained Paddy Disease Classification. *2025 7th International Conference on Applied Computational Intelligence in Information Systems (ACIIS)* (October 2025), 1–6. DOI:https://doi.org/10.1109/aciis66255.2025.11402958

[6] Mingxing Tan and Quoc V Le. 2019. EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. *arXiv.org*. Retrieved from https://arxiv.org/abs/1905.11946

[7] Alhassan Mumuni and Fuseini Mumuni. 2022. Data augmentation: A comprehensive survey of modern approaches. *Array* 16, (November 2022), 100258. DOI:https://doi.org/10.1016/j.array.2022.100258

[8] Mateusz Buda, Atsuto Maki, and Maciej A. Mazurowski. 2018. A systematic study of the class imbalance problem in convolutional neural networks. *Neural Networks* 106, (October 2018), 249–259. DOI:https://doi.org/10.1016/j.neunet.2018.07.011

[9] Shahariar Sarkar. 2025. From Black-Box to Transparent AI in Agriculture: Integrating Grad-CAM and Counterfactual Explanations for Rice Leaf Disease Detection. *2025 7th International Conference on Electrical Information and Communication Technology (EICT)* (December 2025), 1–6. DOI:https://doi.org/10.1109/eict68394.2025.11355586

[10] Anshul M. 2024. Rice Disease Dataset. *Kaggle.com*. Retrieved from https://www.kaggle.com/datasets/anshulm257/rice-disease-dataset

[11] shayanriyaz. 2019. Rice Leafs. *www.kaggle.com*. Retrieved from https://www.kaggle.com/datasets/shayanriyaz/riceleafs

[12] Shruti Agarwal. 2021. Philippines Rice Diseases. *Kaggle.com*. Retrieved from https://www.kaggle.com/datasets/shrupyag001/philippines-rice-diseases

[13] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. 2017. Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. *2017 IEEE International Conference on Computer Vision (ICCV)* (October 2017), 618–626. DOI:https://doi.org/10.1109/iccv.2017.74

[14] Richard S Sutton and Andrew Barto. 2018. *Reinforcement learning: An introduction* (2nd ed.). The Mit Press, Cambridge, Ma ; London.